OKUBE

Network + Energy Aware Scheduler

Okube: extending a custom network-aware orchestrator with carbon-aware placement logic

CUSTOM GO ORCHESTRATOR | MIDSEM EVALUATION | PATH TO FUSED SCHEDULER

Built as a custom scheduler, by design — not a Kubernetes plugin — to keep full control over placement logic.

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Project Status

Current project state

- Manager / worker model
- Docker-based orchestration
- Custom scheduler (no Kubernetes dependency)
- etcd-based coordination

Network-aware scheduling

Already exists in the pipeline — latency and topology cost inform placement decisions today.

Energy-aware extension

Added as a minimal, incremental scoring layer — not a rewrite of the scheduler.

> *Rather than starting from scratch, I extended the scheduler I had already built for this project.*

Research direction: combine network cost and energy cost in one fused scheduling score.

WHAT WE ARE BUILDING ON

Research Basis

NETWORK

Diktyo

Network-aware placement

- Latency- and bandwidth-aware scheduling
- Strong fit for microservice placement

ENERGY

Saad et al. (2025)

Federated Kepler-based energy prediction

- Predicts per-workload energy consumption
- Does not decide scheduling itself

MECHANISM

Piontek et al.

Scheduling mechanism pattern

- Distinguishes critical vs. shiftable jobs
- Delays shiftable work to lower-carbon windows

TAKEAWAY The contribution is not to reproduce one paper wholesale — it is to combine the stronger energy signal (Saad et al.) with the scheduling mechanism (Piontek et al.) on top of the existing network-aware base (Diktyo).

HOW THE FUSED SCORE IS COMPUTED

System Architecture

TASK METADATA

criticality · flexibility · deadline

NODE METADATA

carbon intensity · resource load · topology / latency

ENERGY ESTIMATE

predict energy draw per node / task pair

→

CARBON CONVERSION

energy × carbon intensity = CO_2 cost

→

FUSED SCORE

network score + energy score

DECISION POLICY

Critical jobs

Prefer safe, lower-latency placements — network cost dominates the fused score.

Flexible jobs

Prefer lower-carbon, lower-energy placements — energy cost dominates the fused score.

Evaluation & Future Work

BASELINES UNDER COMPARISON

Configuration	Network-aware	Energy-aware
Default scheduler	–	–
Network-only	✓	–
Energy-only	–	✓
Fused network + energy	✓	✓

Evaluation metrics

metric → how it will be measured

- Total CO_2 emitted
  - predicted energy × node carbon intensity, summed per run
- p95 latency / throughput
  - task start–finish timestamps from the scheduler log
- Deadline violation rate
  - share of tasks completed after their metadata deadline
- Scheduling overhead
  - wall-clock time spent in the scoring path per cycle

Future work

- Run the 4-configuration comparison and report results
  - weights + enable/disable flags
- Real Kepler power model
- Live carbon intensity feed
  - static config in place — swap in a live API
- Stronger task flexibility policy
- Eventual port to a Kubernetes scheduling framework